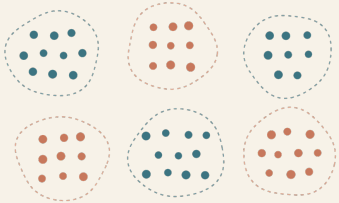
# Adaptive Design of Clustered Surveys and Experiments

Carlos Gonzalez-Perez, University of Oxford



# Outline



Introduction and Motivation

Model and Study Policies

Adaptive Policies

Conditional Universality

Application

Conclusion

# A Survey in Great Britain

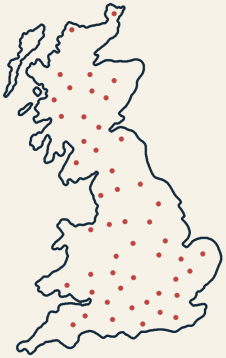




Sedgwick, P. (2015). Multistage sampling

# Clustered Designs

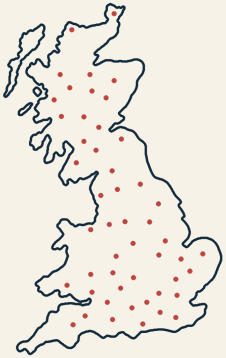


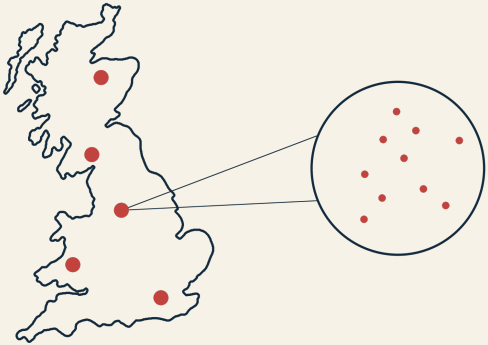


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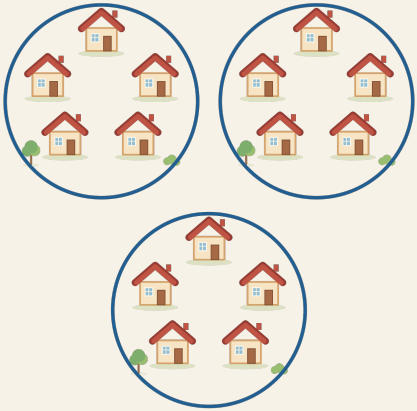




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# More Clusters or More Units per Cluster?





# Optimal Allocation



- **trade-off** between **sampling costs** and **dependence of units within clusters**
- for a given **budget**, the **optimal** (variance minimizing) allocation depends on
  - cost structure, usually **known**
  - **intra-cluster correlation** (ICC), usually **unknown**
- **Guessing is a bad idea:** Knox and Chondros (2004) report variations of the ICC of one order of magnitude within the same population

# Research Questions

- **Can we design optimal clustered surveys and experiments under unknown ICC?**

- **Can we do so without compromising inference?**

# Performance



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  - sample clusters **sequentially** and **learn** the ICC while the experiment runs
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- **Can we design optimal clustered surveys and experiments under unknown ICC?**
- **YES!** Two **adaptive algorithms**
  - sample clusters **sequentially** and **learn** the ICC while the experiment runs
  - implement the **optimal design** in the limit
- **Great properties**
  - **negligible** excess variance
  - **beat any static design** by an order of magnitude
  - **rate unimprovable** (minimax optimal)

# Revisiting Fink, Jack and Masiye (2020, AER)



- **Setting:** effect of easing liquidity constraints on harvest and *ganyu* labor demand
  - favorable loans to farmers in Zambia with treatment at the **village** level
  - 175 villages (clusters) $\times$ 18 households per village (units)

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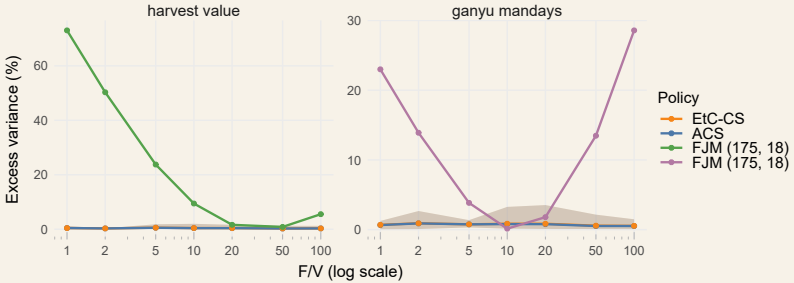


- **Setting:** effect of easing liquidity constraints on harvest and *ganyu* labor demand
  - favorable loans to farmers in Zambia with treatment at the **village** level
  - 175 villages (clusters) $\times$ 18 households per village (units)

- Was the realized design **cost-optimal?**
  - I don't observe the cost ratio $F/V$, but possibly $F/V \in [1, 2]$
  - FJM design only optimal at $F/V \approx 33$ for harvest (and $\approx 12$ for ganyu)
  - away from these ratios, realized design carries **up to 70% excess variance**

# What Would Adaptive Allocation Have Delivered?






*Notes:* Excess variance is presented relative to the oracle design (%). Lines are means over 10,000 simulations, bands are 95% intervals across simulations.

- My algorithms achieve **up to 41%** **lower variance** at equal cost or achieve equal precision at **up to 42%** **lower cost** and are **robust to ICC misspecification**

# Conditional Inference



- **Design depends on sample data**
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  - **without correction nor data exclusion**
- **General principle**
  - Neyman allocations, PropScores-based treatment allocations (Hahn et al., 2011)

# Conditional Inference is Asymptotically Normal



**We were doing fine and we did not know!**

(under error symmetry and policy evenness)

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# Model – Survey Design Problem



- Extends mechanically to experimental design and ATE estimation

- Nested **random effects** model [***A1***]

$$y_{ki} = \mu + u_k + \varepsilon_{ki}, \qquad \mathrm{Var}(u_k) = \tau^2, \quad \mathrm{Var}(\varepsilon_{ki}) = \sigma^2, \qquad u_k \perp\!\!\!\perp \varepsilon_{ki}, \quad iid$$

  - **unknown variance ratio** $\lambda_0 = \sigma^2/\tau^2$, $\mathrm{ICC} = 1/(1+\lambda_0)$
  - $\omega = (\tau^2, \lambda_0) \in \Omega$ with $\Omega$ **known** and **compact** [***A2***], with $\lambda_0 \in [\lambda_m, \lambda_M]$

# Model – Survey Design Problem



- Define the **average of averages** estimator

$$\hat{\mu}_{K,N} = \frac{1}{K}\sum_{k=1}^{K}\frac{1}{N}\sum_{i=1}^{N} y_{ki}, \qquad K \text{ clusters, } N \text{ number of units}$$

- **Goal**

$$\min_{K,N} \ \mathrm{Var}(\hat{\mu}_{K,N}) = W(K,N) \qquad \text{st} \quad K(F+VN) \leq B \ \text{ a.s.}$$

  - **Linear Costs [*A3*]:** $F$ fixed cost/cluster, $V$ variable cost/unit, $B$ fixed budget
  - $W(K,N) = \frac{\tau^2}{K}\left(1+\frac{\lambda_0}{N}\right)$

# Cochran's Allocation and Guessing

- **Oracle** solution goes back to Cochran (1977)

$$N^*(\lambda_0) = \sqrt{\lambda_0 F/V}, \qquad K^*(\lambda_0) = \frac{B}{F + V N^*(\lambda_0)}$$

  - **Optimal map** $\lambda \mapsto (K^*(\lambda), N^*(\lambda))$
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- $\lambda_0$ is **unknown**, so in practice, designers **guess**
  - guessing is **difficult** Knox and Chondros (2004)
  - guessing is **expensive**, as shown in FJM

# Literature Review

- **Survey Theory** *mostly* ignores ICC uncertainty
  - prominent role of ICC, but rarely estimated **inside** the optimization problem
  - algorithmic extensions of Cochran's principles (Bethel, 1989; Ballin and Barcaroli, 2013; Barcaroli et al., 2022)

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  - **Sample-size re-estimation** with nuisance parameters (Stein, 1945)
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- First paper to establish **optimality guarantees** against the oracle under **unknown ICC** in the **joint allocation** problem

# Excess Variance and the Cost of Guessing



- **Oracle variance**

$$W_0 = W(K^*(\lambda_0), N^*(\lambda_0)) = \boldsymbol{\Theta(1/B)}$$

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$$\Delta_\pi = W(K^\pi, N^\pi) - W_0 \geq 0, \qquad \max_{\omega \in \Omega} \mathbb{E}_{\omega,\,\pi}[\Delta_\pi]$$

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- **Cost of guessing** in static policies (Lemma 5.2)

$$\Delta_{\text{guessing}} = W(K^*(\lambda), N^*(\lambda)) - W_0 = \boldsymbol{\Theta(1/B)} \text{ for all } \lambda \neq \lambda_0$$

  - **guessing is expensive**, same order as the oracle variance itself
  - true for **any** static $\lambda$, including minimax $\lambda$ (van Breukelen and Candel, 2015)

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# Pilot-Based Algorithms and Implementability





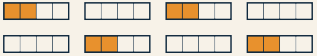
population

# Pilot-Based Algorithms and Implementability





population



pilot $(K_p, N_p) = (4, 2)$

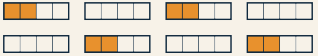
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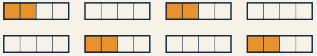
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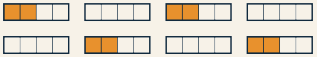
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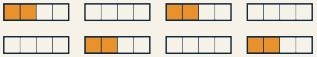
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# Pilot-Based Algorithms and Implementability



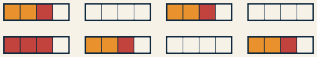


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- What if $K_p > K^*(\hat{\lambda}_p)$? i.e. final design had been $3 \times 4$? **Cluster oversampling**
- **Circularity!** Optimal design requires ICC, but learning ICC requires many clusters

# Escaping the Circularity through Adaptivity



- **Explore-then-Commit Cluster Selection** (`EtC-CS`)
  - **Smallest plausible pilot**

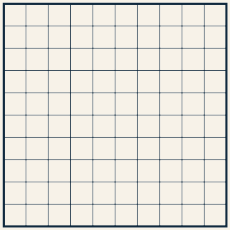
$$(K_p, N_p) = (K^*(\lambda_M), N^*(\lambda_m)) \implies$$
$$K^*(\lambda_M) \le K^*(\hat{\lambda}_p) \text{ and } N^*(\lambda_m) \le N^*(\hat{\lambda}_p) \quad \forall\ \hat{\lambda}_p \in [\lambda_m, \lambda_M]$$

  - Implementable by design/assumption [***A2***]

- `EtC-CS` is already **minimax optimal** (Theorem 5.4), but overly **conservative**

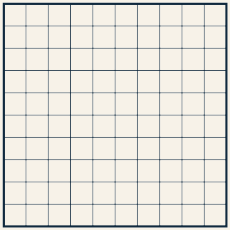
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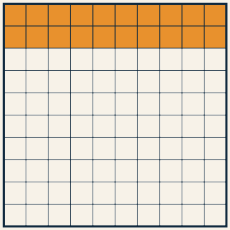




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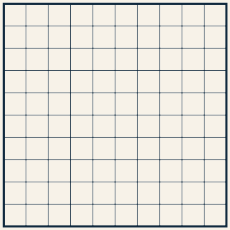


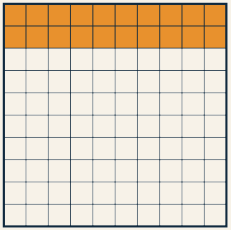


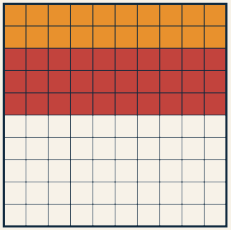


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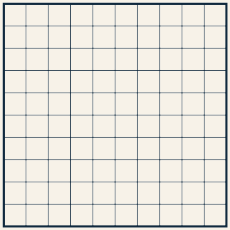


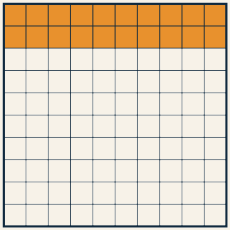


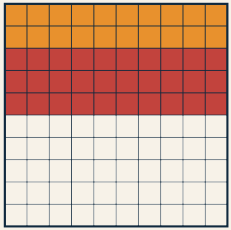


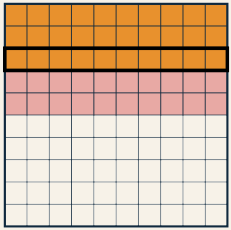
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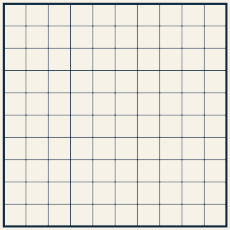


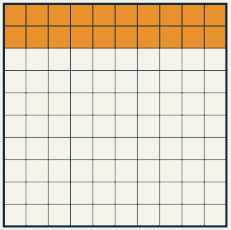


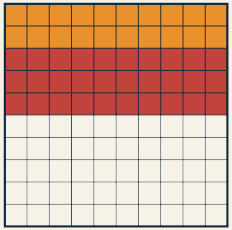


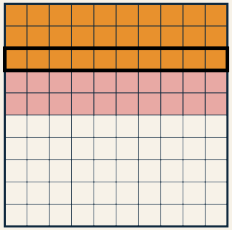
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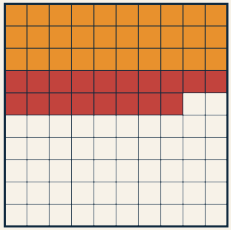












# Adaptive Cluster Selection (ACS)

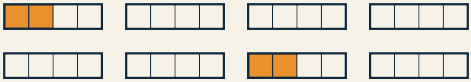


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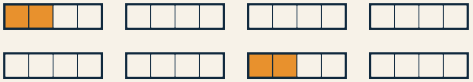
$$(K^*(\hat{\lambda}_{2,2}), N^*(\hat{\lambda}_{2,2})) = (5, 2)$$

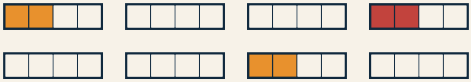
$$2 + 1 = 3 \leq 5 \checkmark$$

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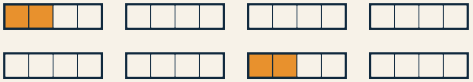
$$(K^*(\hat{\lambda}_{3,2}), N^*(\hat{\lambda}_{3,2})) = (4, 3)$$

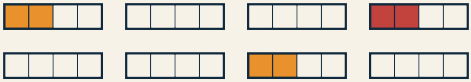
$$3 + 1 = 4 \leq 4 \checkmark$$

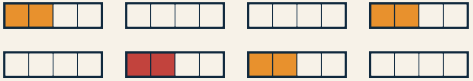
# Adaptive Cluster Selection (ACS)



- fix shallow $N_p = N^*(\lambda_m)$ and **sample one cluster at a time**
- **stop before oversampling** $s = \inf\{k : k + 1 > K^*(\hat{\lambda}_k)\}$







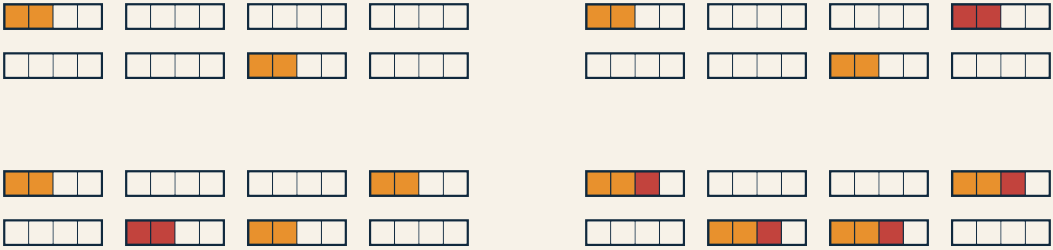
$$(K^*(\hat{\lambda}_{4,2}), N^*(\hat{\lambda}_{4,2})) = (4, 3)$$

$$4 + 1 = 5 \leq 4 \text{ ✗}$$

# Adaptive Cluster Selection (ACS)



- fix shallow $N_p = N^*(\lambda_m)$ and **sample one cluster at a time**
- **stop before oversampling** $s = \inf\{k : k + 1 > K^*(\hat{\lambda}_k)\}$



complete final design

$(K^{\texttt{ACS}}, N^{\texttt{ACS}}) = (4, 3)$

**Theorems 5.4 and 5.5. Upper and Lower Bounds**

Under $[A1 - A4]$ + subgaussian $u_k, \varepsilon_{ki}$ $[A5]$, $\exists$ constants $C_\pi < \infty$ not dep. on $B$

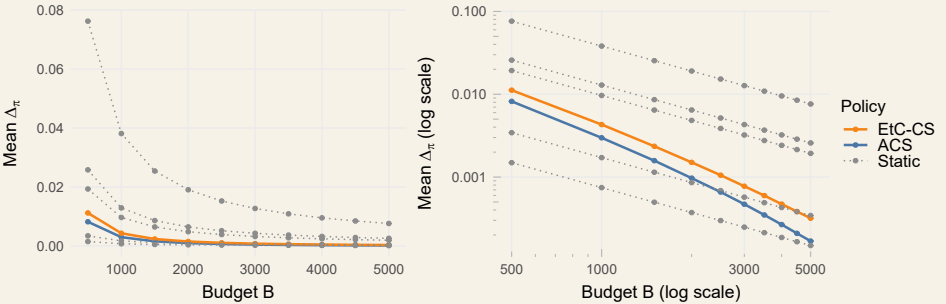
$$\max_{\omega \in \Omega} \mathbb{E}\left[\Delta_{\texttt{Etc-cs}}\right] \leq \frac{C_{\texttt{Etc-cs}}}{B^2}, \qquad \max_{\omega \in \Omega} \mathbb{E}\left[\Delta_{\texttt{ACS}}\right] \leq \frac{C_{\texttt{ACS}}}{B^2}$$

Moreover, for every policy $\pi$ there exists $\omega \in \Omega$ such that $\mathbb{E}[\Delta_\pi] \geq C/B^2$.

- asymptotically **negligible** excess variance, $O(1/B^2) << \Theta(1/B)$
- algorithms asymptotically dominate every **misspecified static policy**
- matching bounds hence **minimax rate optimal**

# Simulations – Excess Variance across Budget

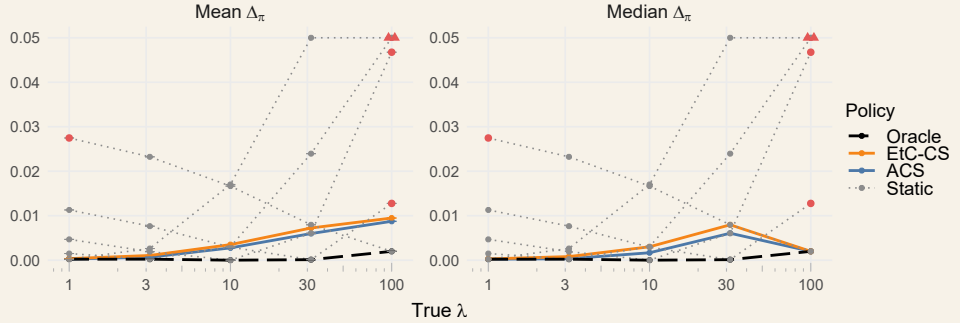





*Notes:* Static policies at $\lambda \in \{1, 3.16, 10, 31.6, 100\}$. $10^5$ replications. $\mu = 0$, $\omega = (1, 5)$, $F = 20$, $V = 2$.

# Simulations – Excess Variance across $\lambda_0$






*Notes:* Static policies at $\lambda \in \{1, 3.16, 10, 31.6, 100\}$. $10^5$ replications. $\mu = 0$, $\tau^2 = 1$, $F = 20$, $V = 2$.

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# Does Adaptivity Compromise Inference? Two Traditions



- $(K^\pi, N^\pi)$ depends on **pilot data** $\Rightarrow$ is $\hat{\mu} \mid (K^\pi, N^\pi)$ centered or asympt normal?
  - adaptive sampling biases estimates and destroys normality
  - (Nie et al., 2018; Chen and Andrews, 2023; Wager, 2024)

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- **Adjustment tradition:** restore normality by **reweighting** the observations
  - Hadad et al. (2021); Zhang et al. (2021); Yin and Shen (2005)

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- **Adjustment tradition:** restore normality by **reweighting** the observations
  - Hadad et al. (2021); Zhang et al. (2021); Yin and Shen (2005)

- **Invariance tradition:** under **exact normality** designs driven by **residuals** do not disturb inference on means
  - Stein (1945); Rinott (1978); Dudewicz and Dalal (1975)

# Inference Invariance under Symmetry



- This paper **invariance tradition**, relaxing **normality** by **error symmetry** (much lighter nonparametric assumption!)

$$(u_k, \varepsilon_{k1}, \varepsilon_{k2}, \dots) \overset{d}{=} -(u_k, \varepsilon_{k1}, \varepsilon_{k2}, \dots) \ \forall\ k \qquad [\boldsymbol{A6}]$$

- **Small cost:** from policies reading residuals, to policies reading **squared errors (even)** or **squared residuals (asymptotically even)**
  - most reasonable policies are even anyway
  - sample variances are functions of squared errors
- Even under gaussianity, **no policy can read levels** like interim means

# Why Symmetry? Signs and Magnitudes



- Under symmetry, cluster errors **factor** into

$$\text{signs} \times \text{magnitudes} \qquad \text{with signs} \perp\!\!\!\perp \text{magnitudes}$$

- Even functions of the data are functions of **magnitudes alone**
  - conditioning on the magnitudes **freezes** the design ...
  - ... but leaves signs **untouched**, so a **CLT can run through them**

# A General Principle



- Mean estimators remain asymptotically **normal** after conditioning on a design
  - generated by any **even** function of the data
  - **arbitrary** sequential designs
  - ICC is a function of variances, hence `EtC-CS` and `ACS` are covered
- Theory **extends** **well beyond** my algorithms and the cluster setting
  - **Neyman allocations** and **optimal propensity scores** (Hahn et al., 2011)
  - allows conditioning on **non-degenerate events** (Hahn et al., 2011)
  - avoids **pilot data exclusion** (Viviano, 2022)

# Asymptotic Evenness



- In practice, policies read **sample** variances
  - demean by the sample mean $\Rightarrow$ designs may depend on the **error signs**

# Asymptotic Evenness



- In practice, policies read **sample** variances
  - demean by the sample mean $\Rightarrow$ designs may depend on the **error signs**

- **Strategy:** nominate an infeasible **surrogate even** policy
  - theory runs through the even surrogate
  - designs of a feasible policy are $p$-close to designs of the even surrogate

# Converging Scale

- **What is close?** Not obvious because $K^*(\lambda_0) = B \cdot h(\lambda_0)$
  - unit $\quad | K^\pi - K^{\tilde{\pi}} | \xrightarrow{p} 0$
  - $o_p(B) \quad | K^\pi/B - K^{\tilde{\pi}}/B | \xrightarrow{p} 0$
  - $o_p(\sqrt{B}) \quad | K^\pi/\sqrt{B} - K^{\tilde{\pi}}/\sqrt{B} | \xrightarrow{p} 0$

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  - $o_p(\sqrt{B}) \quad | \, K^\pi/\sqrt{B} - K^{\tilde{\pi}}/\sqrt{B} \, | \xrightarrow{p} 0$

- **Only $o_p(\sqrt{B})$ and faster are usually feasible**
  - replacing sq. errors by sq. residuals usually **prevents unit coupling**
  - but $o_p(\sqrt{B})$ (and faster $o_p(B)$) remains **feasible**
  - $\pi$ is **asymptotically even** if $| \, K^\pi/\sqrt{B} - K^{\tilde{\pi}}/\sqrt{B} \, | \xrightarrow{p} 0$ and $| \, N^\pi - N^{\tilde{\pi}} \, | \xrightarrow{p} 0$

# Scale of the Conditioning Events



- $A_B$ : $((K^\pi, N^\pi) \in C)$ $((K^\pi/B, N^\pi) \in C)$ $((K^\pi/\sqrt{B}, N^\pi) \in C)$

# Scale of the Conditioning Events



- $A_B$ : $((K^\pi, N^\pi) \in C)$ $((K^\pi/B, N^\pi) \in C)$ $((K^\pi/\sqrt{B}, N^\pi) \in C)$

- **Non-degeneracy of conditioning events**
  - recall $K^*(\lambda_0) = B \cdot h(\lambda_0) \Rightarrow$ consistent designs $K^\pi/B \xrightarrow{p} h(\lambda_0)$
  - events $(K^\pi/B, N^\pi)$ are **degenerate** (conditional = unconditional inf)
  - while events $(K^\pi/\sqrt{B}, N^\pi)$ remain **non-degenerate**, like $K^\pi$ quantiles

# Summary Scale Conditioning Events



| **Scale** | **Feasible Surrogate Coupling?** | **Non-degenerate Events?** | **Tools** |
|---|---|---|---|
| Unit | ✗ | – | – |
| $1/B$ | ✓ | ✗ | Anscombe |
| $1/\sqrt{B}$ (new) | ✓ | ✓ | Symmetry and Evenness |

$\sqrt{B}$ is the *only* feasible scale with non-degenerate events

**Theorem 6.10. Design-Robust Inference for `EtC-CS` and `ACS`**

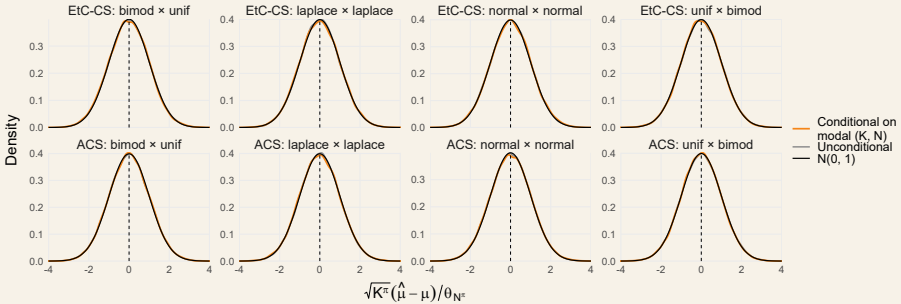
Under $[A1-A4]$ + symmetry $[A6]$, for *any* design event $A_B = ((K^\pi/\sqrt{B}, N^\pi) \in C)$

$$\mathscr{L}\left( \frac{\sqrt{K^\pi}\,(\hat{\mu}_{K^\pi,\,N^\pi} - \mu)}{\hat{b}_{K^\pi,\,N^\pi}} \;\middle|\; A_B \right) \Longrightarrow N(0,1) \quad \pi \in \{\texttt{EtC-CS}, \texttt{ACS}\} \quad \lim_{B\to\infty} \mathbb{P}(A_B) > 0$$

- **Nothing needs to change:** full sample, usual clustered SEs and normal CV—no reweighting, no sample splitting, no discarded data

# Simulations – Inference is Unaffected

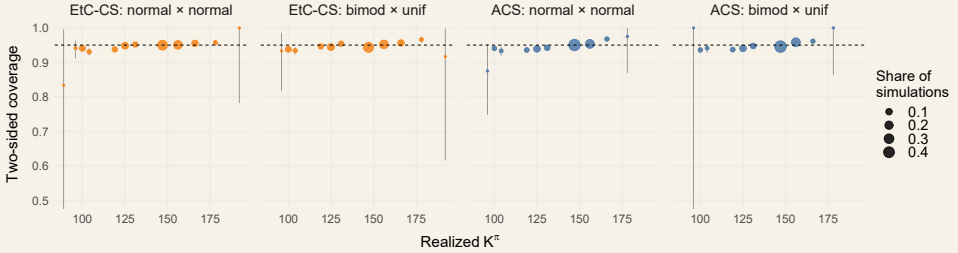





*Notes:* Empirical density at the modal design. Distribution of $u_k$ (left), of $\varepsilon_{ki}$ (right). Distributions considered: Bimodal-Uniform, Laplace-Laplace, Normal-Normal, Uniform-Bimodal. $10^5$ replications. $\mu = 0$, $\omega = (1, 5)$, $B = 2{,}000$, $F = 20$, $V = 2$.

# Simulations – Cell-Conditional Inference

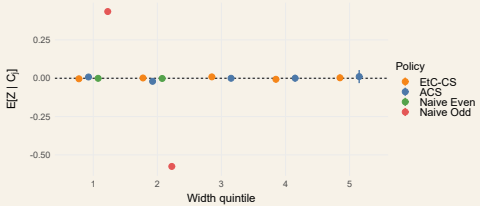




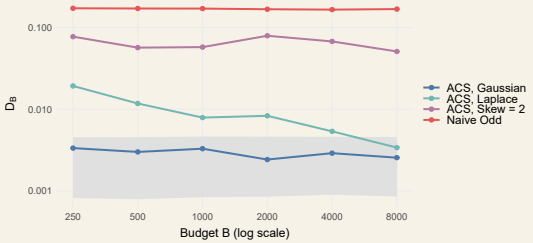

*Notes:* Design-conditional coverage of $\hat{\mu}_{K_B^\pi, N_B^\pi} \pm 1.96 \hat{b}_{K_B^\pi, N_B^\pi} / \sqrt{K_B^\pi}$. Grey lines are Clopper–Pearson 95% intervals for each realized $K_B^\pi$. $10^5$ replications. $\mu = 0$, $\omega = (1, 5)$, $B = 5{,}000$, $F = 20$, $V = 2$.

# Simulations – Sharp Boundary






(a) $E[Z_B^\pi \mid C_j]$




(b) $D_B = \sup_{t \in \mathcal{T}} \max_j \left| \mathbb{E}[e^{itZ_B^\pi} \mathbb{1}_{C_j}] - \phi(t)\mathbb{P}(C_j) \right|$

*Notes:* $(C_j)_j$ are quintiles of the realized $K^\pi$. Grey band represents worst-case 95% range of $D_B$ over $B$ under exact conditional normality. Symmetric errors are $N(0,1)$, asymmetric errors are centered $\Gamma(1,1)$. $10^5$ replications. $\mu = 0$, $\tau^2 = 1$, $B = 2{,}000$, $F = 20$, $V = 2$.

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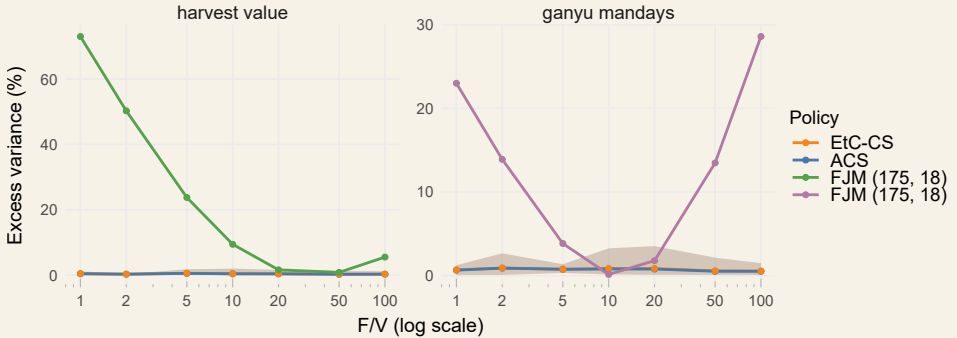
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# Fink et al (2020, AER)



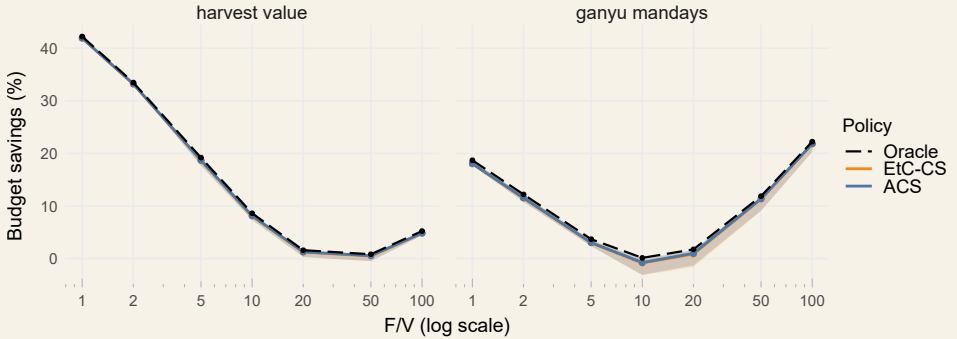



*Notes:* Excess variance is presented relative to the oracle design (%). Lines are means over 10,000 simulations, bands are 95% intervals across simulations.

- Up to 41% (22%) **lower variance** at equal cost

# Counterfactual Savings






*Notes:* Budget savings (%) relative to the imputed budget $B = 175\times(F+17.9V)$. $10^4$ replications. 95% confidence intervals across simulations at selected budget.

- Up to 42% (22%) **lower cost** at equal variance

# Inference and Standard Errors



| | Paper | FJM | FJM | $F/V$ | | | |
|---|---|---|---|---|---|---|---|
| | (Real) | (Real) | (Synthetic) | 1 | 5 | 20 | 100 |
| $K/N$ | 175/17.9 | 175/17.9 | 175/18 | 774/3 | 332/7 | 194/14 | 157/32 |
| Cash | 0.105* | 0.112** | 0.112*** | 0.109*** | 0.114*** | 0.112*** | 0.111*** |
| | (0.041) | **(0.046)** | **(0.042)** | **(0.035)** | (0.039) | (0.042) | (0.040) |
| Maize | 0.056 | 0.037 | 0.036 | 0.034 | 0.037 | 0.037 | 0.038 |
| | (0.054) | **(0.047)** | **(0.042)** | **(0.035)** | (0.039) | (0.042) | (0.040) |

Treatment-effect estimates for log(harvest value + 1), year-one outcomes.

*Notes:* Column 1 is reproduced from Fink et al. (2020), Table 4 Panel B (years one and two pooled). Counterfactual designs are generated by `ACS`; coefficients and standard errors are means over 1,000 simulations, and $K/N$ reports the mean realized design. CR1 cluster standard errors, as in the original. Stars from the normal approximation: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

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# Summary



- **Framework** for cost-optimal clustered designs under **unknown ICC**
- Two **adaptive algorithms**
  - negligible excess variance
  - dominate any misspecified static design, rate unimprovable
- **Adaptivity does not compromise inference** under **error symmetry**
  - asymptotically even designs do not disturb inference
  - no corrections, no discarded data
  - general principle beyond clusters as in neyman allocations
- Economically **meaningful**
  - 41% lower variance at equal cost, 42% lower cost at equal variance
  - runs live in the field (online survey coming up)

# Thank you!

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